

MERA — Multi-channel Deterministic RF Event Recorder IP

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Executive Summary

AMD Versal RF VR1952에서 범용 FFT/channelizer/resampler/AIE interconnect/DataMover를 독립 IP로 대체하는 방향은 차별화 근거가 약하다. 공개 설계에서 반복적으로 나타나는 capture/trigger/deep-capture/data-recording 영역을 상위 application IP인 MERA로 묶는 것이 현재 가장 근거가 좋은 후보이다.

VR1952 기준 사양

항목	VR1952	근거
RF ADC	8 × 14-bit, 32 GSPS	AMD DS950 / product page
RF DAC	16 × 14-bit, 16 GSPS	AMD DS950 / product page
AIE	120 tiles	AMD DS950 / product page
LDPC	6 decoders	AMD DS950 / product page
Channelizer	DS950: 96 × 1 GSPS; web page: 320	문서 정의/버전 차이
FFT/iFFT	DS950: 36; web page: 40	문서 정의/버전 차이
Poly block	24	AMD DS950
LUT	1,130,880	AMD DS950
DSP58	3,976	AMD DS950

AMD가 이미 제공하는 것

- 4 GSPS FFT/iFFT, 8~4096 point, 18-bit input/31-bit output.
- 64-tap + 8-point FFT channelizer.
- Polyphase arbitrary resampler: 256 phases, 17 taps/phase, 4,352-tap prototype; 1 GSPS/instance.
- AIE stream switch: 32-bit AXI4-Stream crossbar, FIFO, arbiter, backpressure.
- PL→AIE 32 GB/s/column, AIE→PL 24 GB/s/column.
- PL→AIE CDC FIFO 12-deep, stream-switch FIFO 4-deep/port, optional 16-deep FIFO.
- AXI DataMover: stream 8~1024b, MM 32~1024b, burst 2~256 beats.

공개 설계에서 확인되는 고객 영역

Xilinx RFSoc-MTS 공개 설계는 3개 ADC 채널의 64 kilosample internal capture, 네 번째 채널의 deep capture, PL-DRAM 장기 저장, trigger, FIFO flush, DMA를 함께 사용한다. AMD/Xilinx RFDC interrupt example은 stimulus/capture register write가 design specific이라고 직접 명시한다.

MERA 제품 정의

Multi-channel Deterministic RF Event Recorder. RF/DSP/AIE stream을 받아 trigger를 기준으로 pre-trigger/post-trigger 구간을 보존하고, channel alignment, 64-bit timestamp, event ID, status를 record header에 넣어 AXI4-MM/DDR로 저장한다.

v1.0 핵심 스펙

항목	기본값
Channels	8
I/Q	16+16 bit complex
AXI-S	128 bit
Timestamp	64 bit
Pre-trigger	64 Ki samples/channel

항목	기본값
Post-trigger	256 Ki samples/channel
Event queue	64
Trigger	external/software/threshold/upstream event
Output	AXI4-MM S2MM
Header	64 bytes
Alignment	0-sample residual target

검증 및 상품화 판단

- 4/8/16 channel, 128/256/512-bit AXIS, 4K~1M sample buffer sweep.
- FPGA에서 LUT/BRAM/URAM/Fmax/latency/sustained DDR BW 측정.
- 100,000 randomized triggers bit-exact golden comparison.
- 상용화 전에는 AMD confidential/early-access IP 존재 여부와 실제 고객 반복 RTL 규모를 별도로 확인해야 한다.

참고자료

- AMD Versal RF Series Product Page — <https://www.amd.com/en/products/adaptive-socs-and-fpgas/versal/rf-series.html>
- AMD DS950 — Versal Architecture and Product Data Sheet — <https://docs.amd.com/api/khub/documents/BY0FMkLH05y85Nwu4GmWsw/content>
- AMD DS945 — Versal XQ Architecture and Product Data Sheet — <https://docs.amd.com/api/khub/documents/wBgkyKdTzQhGgubawOKewA/content>
- AMD AM009 — AI Engine Architecture Manual: Stream Switch Buffering — <https://docs.amd.com/r/en-US/am009-versal-ai-engine/Stream-Switch-Buffering-and-Latency>
- AMD AM009 — AI Engine to PL Interface — <https://docs.amd.com/r/en-US/am009-versal-ai-engine/AI-Engine-to-Programmable-Logic-Interface>
- AMD AM009 — AXI4-Stream Interconnect — <https://docs.amd.com/r/en-US/am009-versal-ai-engine/AXI4-Stream-Interconnect>
- AMD PG022 — AXI DataMover — https://docs.amd.com/r/en-US/pg022_axi_datamover/Features
- AMD/Xilinx RFSoC-MTS GitHub — <https://github.com/Xilinx/RFSoC-MTS>
- AMD/Xilinx RFDC interrupt example — https://github.com/Xilinx/embeddedsw/blob/master/XilinxProcessorIPLib/drivers/rfdc/examples/xrfdc_intr_example.c